

TLDR: Across 60 attribution graphs spanning chemistry, geography, and GEMMA-2-2B and QWEN3-4B, model architecture explains roughly **14× more variance in circuit structure than knowledge domain**. We introduce **traceback graphing**, identify a **"bottleneck tax"** linking bottleneck-layer activation to lower confidence, and show via 80 steering experiments that pathway topology, cross-circuit frequency, and text-level steerability decouple into a **three-tier dissociation**.

Abstract

We characterize the cross-domain organization of factual knowledge circuits in two large language models by analyzing attribution graphs for 60 prompts spanning three knowledge domains: chemical symbol lookup (10 prompts), world capital retrieval (10 prompts), and historical date recall (10 prompts), generated for both GEMMA-2-2B and QWEN3-4B via the Neuronpedia API. We introduce **traceback graphing**; a backward breadth-first search with geometric score decay (factor 0.8) that traces critical paths from output predictions back to input features and identifies bottleneck features where 60% or more of paths converge.

Per-layer activation energy profiling reveals that within-model profile similarity is **0.978** across knowledge domains while between-model similarity is only **0.696**. We term this pattern **"architecture dominance,"** the observation that within-model energy profile similarity clusters roughly 14× more tightly than between-model similarity, with knowledge domain modulating the profile by less than 2%. The two architectures use complementary energy accumulation strategies, GEMMA front-loading 54% of activation energy into layers 0–12 while QWEN back-loads with only 31% in the corresponding first half, yet both achieve comparable factual recall accuracy. Within each architecture we identify **universal bottleneck features** (features serving as bottlenecks across all three knowledge domains): six in GEMMA and 15 in QWEN, with zero overlap between architectures. Their dominant categories (code, language) suggest a routing-infrastructure function rather than a knowledge-storage one.

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In GEMMA, activation energy at the bottleneck layer L6 negatively correlates with output confidence (Spearman $r = -0.684$, Bonferroni-significant), while energy at post-bottleneck layers L13 and L16 positively predicts confidence ($r \approx +0.60$). We term this pattern the "**bottleneck tax**," heavier activation concentration at the bottleneck layer predicting lower downstream confidence, and interpret it through information bottleneck theory (Tishby & Zaslavsky, 2015) as evidence consistent with over-compression at intermediate layers reducing the information available for downstream evidence accumulation. Causal validation via 80 steering experiments reveals a **three-tier dissociation**: essential-pathway topology predicts distributional perturbation strength (mean KL = 1.448), cross-circuit frequency does not predict causal influence, and text-level output changes are governed by output entropy rather than feature-selection criteria. Circuit redundancy (94.1% of nodes are non-essential) absorbs even strong tier-1 perturbations before they reach the output layer.

Key Findings

- **Architecture dominance.** Across 60 circuits, model architecture explains ~14× more variance in layer-energy profiles than knowledge domain. GEMMA front-loads 54% of energy into early layers. Qwen back-loads with 31%. Both

Large language models encode vast factual knowledge, yet how this knowledge is organized at the circuit level remains poorly understood. When a model predicts that the chemical symbol for gold is "Au" or that the capital of Japan is "Tokyo," which internal features are responsible?

A central question is whether different knowledge types (symbolic mappings, geographic relations, temporal facts) use separate specialised circuits or share common computational infrastructure. Both possibilities are consistent with the existing mechanistic-interpretability literature, and the question has been studied only piecemeal. L E F

The answer matters for model editing and targeted intervention. If factual knowledge is organized by domain, interventions must be domain-specific. If circuits share universal bottleneck features^①, a small number of features could provide efficient intervention points across multiple knowledge types.

1.2 Research Questions

- ① **Within-domain convergence.** Do prompts querying the same type of knowledge (e.g., chemical symbols) share circuit architecture?
- ② **Cross-domain divergence.** Do different knowledge types use different circuit features or layer depths?
- ③ **Universal bottlenecks.** Are there features that participate in circuits across all knowledge domains? → Universal features
- ④ **Architecture vs. domain.** Is circuit structure more determined by model architecture or by the type of knowledge being retrieved?

1.3 Contributions

We present six contributions:

1. **Systematic cross-domain circuit analysis.** The first large-scale comparison of attribution circuits across three knowledge domains on two model architectures (60 circuits total), with statistical validation via permutation tests ($p < 0.0001$).
2. **Architecture-dominance finding.** Demonstration that bottleneck depth is determined by model architecture, not knowledge domain. GEMMA consistently uses early bottlenecks (L5-7) and Qwen uses late bottlenecks (L22-25).

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Fancy methods

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following + maybe?

"Bottleneck, new key features?"

- ① **Systematic cross-domain circuit analysis.** The first large-scale comparison of attribution circuits across three knowledge domains on two model architectures (60 circuits total), with statistical validation via permutation tests ($p < 0.0001$).
- ② **Architecture-dominance finding.** Demonstration that bottleneck depth is determined by model architecture, not knowledge domain. GEMMA consistently uses early bottlenecks (L5–7) and QWEN uses late bottlenecks (L22–25) regardless of whether the task involves chemistry, geography, or history. Cross-model Mann–Whitney tests confirm 13/16 structural metrics differ significantly (Bonferroni-corrected).
- ③ **Universal bottleneck identification.** Discovery of 6 universal bottleneck features in GEMMA and 15 in QWEN that participate in circuits across all three knowledge domains, with zero overlap between architectures. Central?
- ④ **Output-path convergence dissociation.** Finding that same-domain circuits share output features far more than intermediate features (up to 4.4× ratio), revealing convergent predictions from divergent internal processing.
- ⑤ **Confidence-energetics model.** A three-predictor regression model explains 49% of GEMMA prediction-confidence variance using total activation energy^①, edge weight kurtosis, and circuit size ($F(3,26) = 8.30, p < 0.001$). Bottleneck position and convergence do not significantly predict confidence, constraining "bottleneck quality" explanations.
- ⑥ **Three-tier causal dissociation.** 80 steering experiments reveal that essential-pathway features produce strong distributional perturbations (mean KL = 1.448) but neither pathway topology nor cross-circuit frequency predicts text-level output changes: circuit redundancy absorbs perturbations, and output determinism governs steering susceptibility.

2. Methodology

Attribution graphs were generated via the Neuronpedia

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Attribution graphs were generated via the Neuronpedia `/api/graph/generate` endpoint with `maxFeatureNodes=3000` , `desiredLogitProb=0.95` , `nodeThreshold=0.8` , `edgeThreshold=0.85` . Graphs typically contain 1,000–1,500 nodes and 40,000–80,000 edges. Our core analytical method is a backward breadth-first search from the top-K output nodes that traces critical paths through intermediate layers and identifies bottleneck features where 60%+ of paths converge.

reference → Where does path converge
recons

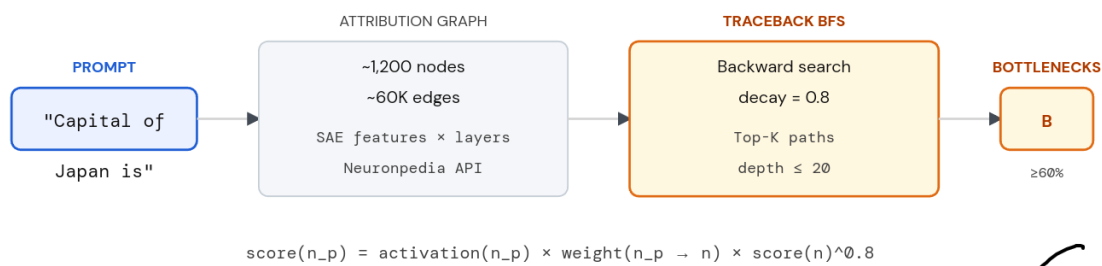


Figure 1. Traceback graphing pipeline. ^① Starting from the highest-contributing output nodes, the algorithm walks backward through the attribution graph, scoring predecessors by activation × edge weight × decayed downstream score. Geometric decay (factor 0.8) prevents score overflow across 20+ layers while preserving relative path ranking.

Models & SAEs

- REFS.
- **GEMMA-2-2B** (Google): 26 layers, ~2B parameters, `gemmascope-transcoder-16k` SAE (16,384 features per layer)
 - **QWEN3-4B** (Alibaba): 36 layers, ~4B parameters, `transcoder-hp` SAE

Prompt Design

Three knowledge domains × 10 prompts each, with template-controlled phrasing:

Domain	Template	Examples
Chemistry	"The chemical symbol for [element] is"	gold → Au, iron → Fe,